FNCE30007 Mid Sem Practice Questions 2024 S1

Derivative Securities (University of Melbourne)

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Question 1

Which of the following is true?

a) Forward and futures contracts are marked to market.
b) Neither futures contracts nor forward contracts are traded on exchanges.
c) Both forward and futures contracts are traded on exchanges.
d) Forward contracts are traded on exchanges, but futures contracts are not.
e) None of the above.

Question 2

In the wheat futures contract, different grades of wheat can be delivered (with price adjustments specified by the exchange) and there are several delivery locations. Which of the following statements is true, ceteris paribus (all other things being equal)?

a) This flexibility tends increase the futures price.
b) This flexibility tends decrease the futures price.
c) This flexibility may increase and may decrease the futures price.
d) This has no effect on the futures price.
e) None of the above.

Question 3

A company shorts a futures contract for 10,000 units of a commodity at $1.50 per unit. The initial margin is $4,500 and the maintenance margin is $3,000. What futures price per unit will result in a margin call?

a) $1.45 per unit
b) $1.55 per unit
c) $1.40 per unit
d) $1.36 per unit
e) None of the above.

Question 4

The following details are to be used for **Questions 4 and 5**. Suppose the three-year interest rates in Australia and Japan are 5% p.a and 6.5% p.a (continuously compounded) respectively. The spot exchange rate is 1 AUD = 1.2540 JPY. The no arbitrage three year forward rate to 4 decimal places is

a) 1 AUD = 1.1988 JPY
b) 1 AUD = 1.2169 JPY
c) 1 AUD = 1.3117 JPY
d) 1 AUD = 1.2606 JPY
e) None of the above.

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Question 5

Refer to the details in **Question 4.** Assume the observed 3 year forward rate is 1AUD = 1.2860 JPY. Using an initial borrowing of 1000 units (in the relevant currency) an appropriate arbitrage strategy to 2 decimal places is

a) Borrow 1000JPY and convert to 797.45AUD. Take a short forward position to sell 926.50 AUD for 1191.48 JPY.
b) Borrow 1000JPY and convert to 797.45AUD. Take a short position to sell 894.44AUD for 1150.25 JPY.
c) Borrow 1000AUD and convert to 1254 JPY. Take a long forward position to buy 1161.83 AUD for 1524.00 JPY
d) Borrow 1000AUD and convert to 1254 JPY. Take a long forward position to buy 1161.83 AUD for 1494.12 JPY.
e) None of the above.

Question 6

Which of the following statements is true holding everything else equal?

a) The longer the time to maturity, the greater the value of a European call.
b) An American put is worth less when the underlying stock pays dividends over the life of the option.
c) A decrease in volatility expectations will reduce the likelihood that an in the money American put on a non-dividend paying stock will be exercised early.
d) A higher risk-free rate will always decrease the value of a European call.
e) None of the above.

Question 7

Consider a European call and put option on an underlying stock currently trading at $30. Both options have a maturity of 6 months and a strike of $28. The put is priced at $3 and the call is priced at $5. The risk-free rate is 6% p.a continuously compounded. Which of the following statements is true to 4 decimal places?

a) There is no arbitrage opportunity
b) An arbitrageur can sell the call and bonds, buy a put and the stock, and immediately receive a riskless profit of $0.8275.
c) An arbitrageur can buy the call and bonds, sell a put and the stock, and immediately receive a riskless profit of $0.8275.
d) None of the above

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Question 8

Which of the following statements is false?

a) An increase in expected volatility of the underlying asset will increase the value of all options (on the underlying asset) given their asymmetric payoffs.
b) The time value of an option can be greater than its intrinsic value.
c) An American option will not always be worth the same or more than an otherwise equivalent European call option.
d) You should never exercise early an American call option on a non-dividend paying stock.
e) None of the above.

Question 9

Consider an equity portfolio with a beta of 0.8 currently valued at $10,000,000. The fund manager seeks to increase the risk of the fund over the next 2 weeks by increasing the beta to 1.35. The index futures are trading at 5,500 points. Each contract point is worth $15. The fund manager should (to the nearest integer)

(a) Long 67 contracts
(b) Long 24 contracts
(c) Short 24 contracts
(d) Short 67 contracts
(e) None of the above

Question 10

Suppose the one-year interest rates in Australia and Canada are respectively 3.5% p.a and 5% p.a continuously compounded. The spot exchange rate is 1AUD = 1.1154CAD. The one year forward rate (to 4 decimal places) is

(a) 1AUD = 1.0988 CAD
(b) 1AUD = 1.0621 CAD
(c) 1AUD = 1.1277 CAD
(d) 1AUD = 1.1323 CAD
(e) None of the above.

Question 11

Consider a forward contract for a coupon bond. The spot price for the bond is currently $950. The forward contract has two years to maturity and coupon payments of $100 are due in 6 months and 18 months' time. Continuously compounded risk-free interest rates % p.a are

Period	Rate (%)
1 month	3.50
6 months	4.00

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12 months	4.50
18 months	5.00
24 months	6.00

The forward price to 2 decimal places should be

(a) $858.66
(b) $856.00
(c) $971.41
(d) $855.54
(e) None of the above

Question 12

Consider an American put option on a non-dividend paying stock. The stock is currently priced at $60 and the put is priced at $3. The option has 6 months to expiration and a strike of $66. The risk-free rate is 8% p.a. continuously compounded and the market expects it to rise to 9% p.a over the coming week. Which of the following statements is true?

(a) An American put should never be exercised early.
(b) An arbitrageur can buy the put and the stock, and exercise immediately, earning a profit of $3.
(c) An arbitrageur can buy the put and the stock and exercise immediately, earning a profit of $2.8824.
(d) An arbitrageur can sell the put and stock and buy bonds, immediately earning a profit of $2.8824.
(e) None of the above.

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